Supplementary Information: Prevalence-adaptive F-measure thresholding for imbalanced tabular classification under label shift

S1 Implementation details

Base learners (identical for all datasets). LightGBM: up to 2,000 trees, learning rate 0.05, 31 leaves, row and column subsampling 0.8, minimum 20 samples per leaf, early stopping after 50 rounds on a 10% slice of the training split. XGBoost: histogram method, depth 6, learning rate 0.05, subsampling 0.8, up to 2,000 trees, early stopping after 50 rounds. Logistic regression: L2 penalty with C = 1, at most 300 L-BFGS iterations. Perceptron: 256 and 128 ReLU units, dropout 0.1, AdamW (learning rate 0.001, weight decay 1e-5), batch size 512, at most 40 epochs, patience 5.

Pre-processing. Categorical features are ordinal-encoded for tree models and frequency-encoded for the linear and neural models; the latter also receive median imputation and standardisation. All statistics are fitted on the training split.

Decision layer. Candidate thresholds: validation scores of each class, thinned to at most 1,024 class-wise quantiles. Histogram: $B = \min(32, \max(8, \lfloor n1/10 \rfloor))$ class-balanced quantile bins, Laplace smoothing 0.5. Prevalence grid: 161 points uniform in $\mathrm{logit}(\pi 0) \pm 9$. Prior: $\sigma = 2$. Guard level: 0.01. EM baseline: at most 200 iterations, tolerance 1e-7, on isotonic-calibrated posteriors.

Random seeds. Splits and learners use seeds 0–4. Static-Bag thresholds are computed on candidate thresholds thinned to 512 quantiles per class. Deployment batches use a generator seeded by a CRC-32 hash of (dataset, model, seed, condition, metric, ρ, m), so every result is exactly reproducible; a complete re-run of the main experiment reproduced all 90,720 result rows with zero difference.

Hardware and software. 4 CPU cores, 15 GB RAM, no GPU; Python 3.13, scikit-learn 1.9.1, LightGBM 4.7.0, XGBoost 3.4.1, PyTorch 2.15.0 (CPU). The device (CPU, CUDA or Apple MPS) is detected automatically.

S2 Results by base learner

Table S1 Mean F1 under shift ($\rho \neq 1$) by base learner

method	lgbm	logreg	mlp	xgb
Default 0.5	0.406	0.342	0.367	0.414
Static-F	0.496	0.447	0.466	0.503
Static-Bag	0.497	0.448	0.466	0.503
EM-0.5	0.449	0.375	0.404	0.459
EM-Fixed	0.507	0.454	0.476	0.516
PrevMatch	0.509	0.452	0.473	0.515
DTA-EM	0.533	0.483	0.501	0.539
PAFT-Plug	0.526	0.475	0.494	0.532
PAFT-Bayes	0.532	0.481	0.500	0.538
PAFT-Guard	0.531	0.480	0.499	0.537
Oracle-π	0.537	0.487	0.505	0.543
Oracle-threshold	0.565	0.515	0.533	0.571

S2 (continued)

Table S1b Paired tests of PAFT-Bayes by base learner (unit: dataset)

model	baseline	n_units	mean_diff	ci95_lo	ci95_hi	wins	ties	losses	p_wilcoxon	rank_biserial	p_holm
lgbm	Static-F	18	+0.0355	+0.0270	+0.0453	18	0	0	0.00001	1.000	0.00002
lgbm	DTA-EM	18	−0.0014	−0.0067	+0.0020	14	0	4	0.09874	0.450	0.09874
lgbm	EM-Fixed	18	+0.0255	+0.0178	+0.0337	18	0	0	0.00001	1.000	0.00002
xgb	Static-F	18	+0.0346	+0.0232	+0.0456	17	0	1	0.00011	0.930	0.00021
xgb	DTA-EM	18	−0.0015	−0.0071	+0.0020	15	0	3	0.03036	0.579	0.03036
xgb	EM-Fixed	18	+0.0220	+0.0159	+0.0280	18	0	0	0.00001	1.000	0.00002
logreg	Static-F	18	+0.0342	+0.0253	+0.0425	17	0	1	0.00002	0.988	0.00003
logreg	DTA-EM	18	−0.0014	−0.0081	+0.0027	13	0	5	0.05386	0.520	0.05386
logreg	EM-Fixed	18	+0.0274	+0.0209	+0.0338	18	0	0	0.00001	1.000	0.00002
mlp	Static-F	18	+0.0341	+0.0231	+0.0444	17	0	1	0.00033	0.883	0.00066
mlp	DTA-EM	18	−0.0015	−0.0080	+0.0023	14	0	4	0.04828	0.532	0.04828
mlp	EM-Fixed	18	+0.0239	+0.0189	+0.0288	18	0	0	0.00001	1.000	0.00002

S3 Batch size

Table S2 Mean F1 under shift by batch size (LightGBM)

method	100	500	2000	10000
Default 0.5	0.399	0.405	0.407	0.408
Static-F	0.499	0.496	0.497	0.497
Static-Bag	0.499	0.496	0.498	0.497
EM-0.5	0.447	0.449	0.450	0.451
EM-Fixed	0.484	0.503	0.508	0.511
PrevMatch	0.476	0.497	0.512	0.517
DTA-EM	0.525	0.530	0.534	0.536
PAFT-Plug	0.495	0.517	0.528	0.533
PAFT-Bayes	0.517	0.527	0.534	0.535
PAFT-Guard	0.518	0.528	0.533	0.532
Oracle-π	0.536	0.536	0.538	0.538
Oracle-threshold	0.628	0.583	0.562	0.551

S4 Seed variability

Table S3 Mean F1 under shift: mean, standard deviation, minimum and maximum over the five seeds (each seed value is a mean over 18 datasets)

method	mean	sd_over_seeds	min	max
Default 0.5	0.406	0.0063	0.396	0.412
Static-F	0.496	0.0017	0.494	0.499
Static-Bag	0.497	0.0020	0.494	0.499
EM-0.5	0.449	0.0060	0.441	0.456
EM-Fixed	0.507	0.0072	0.499	0.517
PrevMatch	0.509	0.0036	0.503	0.512
DTA-EM	0.533	0.0024	0.530	0.536
PAFT-Plug	0.526	0.0044	0.519	0.530
PAFT-Bayes	0.532	0.0029	0.527	0.534
PAFT-Guard	0.531	0.0021	0.528	0.533
Oracle-π	0.537	0.0021	0.534	0.539
Oracle-threshold	0.565	0.0019	0.563	0.567

S5 Real suites: all rules including the exploratory sequential variant

Table S4 Mean F1 on real-shift suites. PAFT-Seq centres the prior on the previous batch's posterior mean ($\sigma = 1$); it was not part of the pre-specified comparison

method	acs_income_states	bank_time	creditcard_time	electricity_time	elliptic_time	kdd99_time
Default 0.5	0.714	0.000	0.645	0.449	0.483	0.973
Static-F	0.702	0.233	0.645	0.450	0.483	0.973
Static-Bag	0.702	0.234	0.644	0.451	0.483	0.975
EM-0.5	0.707	0.265	0.636	0.423	0.477	0.947
EM-Fixed	0.692	0.265	0.551	0.424	0.478	0.960
PrevMatch	0.708	0.263	0.635	0.384	0.481	0.973
DTA-EM	0.692	0.265	0.633	0.425	0.484	0.947
PAFT-Plug	0.693	0.265	0.638	0.426	0.477	0.947
PAFT-Bayes	0.693	0.265	0.641	0.426	0.477	0.947
PAFT-Guard	0.702	0.233	0.645	0.450	0.483	0.973
PAFT-Seq (exploratory)	0.693	0.265	0.643	0.426	0.477	0.947
Oracle-π	0.713	0.260	0.646	0.450	0.480	0.950
Oracle-threshold	0.720	0.272	0.729	0.721	0.600	0.988

S5 (continued)

Table S4b Standard deviation over seeds of the values in Table S4

method	acs_income_states	bank_time	creditcard_time	electricity_time	elliptic_time	kdd99_time
Default 0.5	0.001	0.000	0.075	0.008	0.011	0.007
Static-F	0.003	0.034	0.075	0.007	0.011	0.007
Static-Bag	0.003	0.031	0.072	0.008	0.011	0.005
EM-0.5	0.001	0.000	0.069	0.009	0.004	0.034
EM-Fixed	0.004	0.000	0.228	0.006	0.004	0.012
PrevMatch	0.002	0.004	0.073	0.016	0.011	0.007
DTA-EM	0.003	0.000	0.072	0.008	0.019	0.034
PAFT-Plug	0.003	0.000	0.079	0.008	0.004	0.027
PAFT-Bayes	0.003	0.000	0.074	0.008	0.005	0.027
PAFT-Guard	0.003	0.034	0.075	0.007	0.011	0.007
PAFT-Seq (exploratory)	0.003	0.000	0.071	0.008	0.005	0.027
Oracle-π	0.001	0.004	0.069	0.008	0.006	0.029
Oracle-threshold	0.001	0.001	0.067	0.004	0.014	0.001

S5 (continued)

Table S4c Real-shift suites by base learner

suite	model	DTA-EM	PAFT-Bayes	PAFT-Guard	Static-F
acs_income_states	lgbm	0.692	0.693	0.702	0.702
acs_income_states	logreg	0.655	0.656	0.663	0.664
acs_income_states	mlp	0.668	0.669	0.677	0.678
acs_income_states	xgb	0.692	0.693	0.701	0.702
bank_time	lgbm	0.265	0.265	0.233	0.233
bank_time	logreg	0.244	0.243	0.198	0.198
bank_time	mlp	0.260	0.015	0.015	0.015
bank_time	xgb	0.265	0.265	0.221	0.221
creditcard_time	lgbm	0.633	0.641	0.645	0.645
creditcard_time	logreg	0.535	0.551	0.555	0.555
creditcard_time	mlp	0.725	0.717	0.723	0.723
creditcard_time	xgb	0.726	0.742	0.732	0.732
electricity_time	lgbm	0.425	0.426	0.450	0.450
electricity_time	logreg	0.321	0.340	0.463	0.463
electricity_time	mlp	0.310	0.330	0.406	0.406
electricity_time	xgb	0.414	0.416	0.449	0.449
elliptic_time	lgbm	0.484	0.477	0.483	0.483
elliptic_time	logreg	0.291	0.293	0.305	0.305
elliptic_time	mlp	0.272	0.202	0.358	0.361
elliptic_time	xgb	0.481	0.480	0.486	0.486
kdd99_time	lgbm	0.947	0.947	0.973	0.973
kdd99_time	logreg	0.937	0.889	0.941	0.941
kdd99_time	mlp	0.926	0.906	0.936	0.936
kdd99_time	xgb	0.954	0.942	0.973	0.973

S6 Guard firing rate by dataset

Table S5 Fraction of batches under pure label shift on which the guard rejects ($\alpha = 0.01$, LightGBM, all batch sizes)

dataset	guard_fire_rate_under_pure_label_shift
Diabetes130US	0.0003
online_shoppers	0.0003
porto_seguro	0.0006
bank_marketing	0.0010
KDD09_churn	0.0016
default_credit	0.0019
kick	0.0032
KDD09_upselling	0.0054
Click_small	0.0073
census_income	0.0130
adult	0.0184
Amazon_employee	0.0194
nomao	0.0378
MiniBooNE	0.0394
jm1	0.0533
KDD09_appetency	0.0619
APSFailure	0.2044
creditcard	1.0000

S7 Robustness: paired tests

Table S6 Paired tests of PAFT-Bayes under each robustness condition (unit: dataset)

condition	baseline	n_units	mean_diff	ci95_lo	ci95_hi	wins	ties	losses	p_wilcoxon	rank_biserial	p_holm
calib_platt	static	18	+0.0369	+0.0287	+0.0465	18	0	0	0.0000	1.000	0.0000
calib_platt	dta_em	18	+0.0025	−0.0015	+0.0081	10	0	8	0.3466	0.263	0.3466
calib_platt	paft_guard	18	+0.0016	+0.0003	+0.0032	14	1	3	0.0010	0.908	0.0020
noise_0.1	static	18	−0.0045	−0.0342	+0.0209	12	0	6	0.4683	0.205	0.9366
noise_0.1	dta_em	18	+0.0042	−0.0030	+0.0121	14	0	4	0.0432	0.544	0.1295
noise_0.1	paft_guard	18	−0.0173	−0.0432	+0.0037	12	0	6	0.9323	0.029	0.9366
noise_0.25	static	18	−0.0149	−0.0647	+0.0205	13	0	5	0.3927	0.240	0.7855
noise_0.25	dta_em	18	+0.0024	−0.0036	+0.0087	12	0	6	0.1674	0.380	0.5021
noise_0.25	paft_guard	18	−0.0255	−0.0733	+0.0066	13	0	5	0.4683	0.205	0.7855
noise_0.5	static	18	−0.0119	−0.0544	+0.0175	12	0	6	0.3466	0.263	0.6931
noise_0.5	dta_em	18	+0.0027	−0.0041	+0.0098	12	0	6	0.1964	0.357	0.5892
noise_0.5	paft_guard	18	−0.0191	−0.0602	+0.0086	11	0	7	0.7337	0.099	0.7337
valfrac_0.1	static	18	+0.0337	+0.0228	+0.0442	17	0	1	0.0003	0.895	0.0008
valfrac_0.1	dta_em	18	−0.0044	−0.0111	−0.0006	3	0	15	0.0104	−0.673	0.0208
valfrac_0.1	paft_guard	18	−0.0010	−0.0048	+0.0014	11	4	3	0.1401	0.448	0.1401
valfrac_0.25	static	18	+0.0333	+0.0231	+0.0436	17	0	1	0.0001	0.953	0.0002
valfrac_0.25	dta_em	18	−0.0026	−0.0077	+0.0002	8	0	10	0.3466	−0.263	0.3466
valfrac_0.25	paft_guard	18	−0.0004	−0.0033	+0.0015	10	5	3	0.0546	0.604	0.1092

S8 Real suites: paired tests

Table S7 Paired tests of PAFT-Bayes on real suites (unit: deployment batch, averaged over seeds)

suite	baseline	n_units	mean_diff	ci95_lo	ci95_hi	wins	ties	losses	p_wilcoxon	rank_biserial	p_holm
acs_income_states	static	50	−0.0093	−0.0116	−0.0068	3	3	44	0.0000	−0.904	0.0000
acs_income_states	dta_em	50	+0.0014	+0.0008	+0.0020	36	0	14	0.0000	0.678	0.0000
acs_income_states	em_fixed	50	+0.0011	+0.0002	+0.0019	32	0	18	0.0035	0.467	0.0071
acs_income_states	paft_guard	50	−0.0087	−0.0109	−0.0061	3	4	43	0.0000	−0.900	0.0000
acs_income_states	paft_plug	50	+0.0001	+0.0000	+0.0003	9	39	2	0.0329	0.727	0.0329
bank_time	static	10	+0.0321	+0.0135	+0.0545	9	0	1	0.0059	0.927	0.0293
bank_time	dta_em	10	+0.0000	+0.0000	+0.0000	0	10	0	1.0000	0.000	1.0000
bank_time	em_fixed	10	+0.0000	+0.0000	+0.0000	0	10	0	1.0000	0.000	1.0000
bank_time	paft_guard	10	+0.0321	+0.0135	+0.0545	9	0	1	0.0059	0.927	0.0293
bank_time	paft_plug	10	+0.0000	+0.0000	+0.0000	0	10	0	1.0000	0.000	1.0000
creditcard_time	static	14	−0.0044	−0.0179	+0.0129	3	5	6	0.2604	−0.422	1.0000
creditcard_time	dta_em	14	+0.0075	−0.0078	+0.0251	7	1	6	0.6002	0.165	1.0000
creditcard_time	em_fixed	14	+0.0900	+0.0571	+0.1190	11	0	3	0.0017	0.886	0.0085
creditcard_time	paft_guard	14	−0.0044	−0.0179	+0.0129	3	5	6	0.2604	−0.422	1.0000
creditcard_time	paft_plug	14	+0.0030	−0.0054	+0.0127	4	7	3	0.7353	0.143	1.0000
electricity_time	static	31	−0.0244	−0.0341	−0.0156	5	0	26	0.0000	−0.891	0.0000
electricity_time	dta_em	31	+0.0004	−0.0006	+0.0015	13	11	7	0.1790	0.343	0.3579
electricity_time	em_fixed	31	+0.0015	+0.0007	+0.0025	18	6	7	0.0035	0.668	0.0105
electricity_time	paft_guard	31	−0.0244	−0.0341	−0.0156	5	0	26	0.0000	−0.891	0.0000
electricity_time	paft_plug	31	+0.0002	−0.0004	+0.0008	7	20	4	0.5337	0.212	0.5337
elliptic_time	static	15	−0.0053	−0.0268	+0.0148	4	5	6	0.4446	−0.273	1.0000
elliptic_time	dta_em	15	−0.0067	−0.0214	+0.0025	4	7	4	0.7794	−0.111	1.0000
elliptic_time	em_fixed	15	−0.0008	−0.0032	+0.0024	2	7	6	0.2076	−0.500	0.8986
elliptic_time	paft_guard	15	−0.0053	−0.0268	+0.0148	4	5	6	0.4446	−0.273	1.0000
elliptic_time	paft_plug	15	+0.0005	+0.0000	+0.0015	2	13	0	0.1797	1.000	0.8986
kdd99_time	static	14	−0.0263	−0.0523	−0.0060	3	0	11	0.0245	−0.676	0.1227
kdd99_time	dta_em	14	+0.0005	−0.0153	+0.0169	5	1	8	0.5525	−0.187	1.0000
kdd99_time	em_fixed	14	−0.0127	−0.0284	+0.0020	4	1	9	0.1730	−0.429	0.5189
kdd99_time	paft_guard	14	−0.0263	−0.0523	−0.0060	3	0	11	0.0245	−0.676	0.1227
kdd99_time	paft_plug	14	+0.0000	+0.0000	+0.0000	0	14	0	1.0000	0.000	1.0000

S9 Error analysis by shift level

Table S8 Per-batch win and loss rates of PAFT-Bayes against Static-F by odds multiplier

rho	F1_static	F1_paft	win	loss	worst_decile_gain	regret
0.125	0.268	0.315	0.688	0.163	−0.015	0.066
0.250	0.354	0.376	0.644	0.216	−0.022	0.052
0.500	0.446	0.451	0.510	0.294	−0.019	0.036
1.000	0.528	0.524	0.149	0.219	−0.011	0.028
2.000	0.594	0.603	0.633	0.208	−0.007	0.020
4.000	0.642	0.683	0.903	0.049	0.000	0.015
8.000	0.674	0.764	0.973	0.012	0.017	0.011

S9 (continued)

Table S9 Per-dataset gains, relative prevalence error and regret ($\rho \neq 1$)

dataset	gain_vs_static	gain_vs_dta	rel_err_pi	regret	prevalence_%	AUC_mean
creditcard	0.002	−0.042	0.213	0.088	0.173	0.926
KDD09_appetency	0.016	0.002	0.476	0.054	1.780	0.800
KDD09_churn	0.017	0.007	0.290	0.026	7.344	0.690
nomao	0.020	0.000	0.014	0.008	28.562	0.995
KDD09_upselling	0.023	0.001	0.082	0.030	7.364	0.860
porto_seguro	0.025	0.003	0.452	0.030	3.645	0.633
APSFailure	0.028	−0.008	0.070	0.069	1.809	0.990
online_shoppers	0.028	−0.001	0.077	0.032	15.475	0.928
MiniBooNE	0.030	0.000	0.015	0.010	28.062	0.984
bank_marketing	0.032	0.000	0.064	0.026	11.698	0.934
Amazon_employee	0.036	0.003	0.182	0.044	5.789	0.727
Click_small	0.038	0.004	0.129	0.023	16.842	0.698
jm1	0.042	0.001	0.172	0.031	19.348	0.721
Diabetes130US	0.047	0.002	0.158	0.027	11.160	0.672
census_income	0.052	0.001	0.062	0.039	6.206	0.952
default_credit	0.054	0.003	0.073	0.019	22.120	0.779
adult	0.062	−0.000	0.028	0.018	23.928	0.926
kick	0.089	0.000	0.071	0.027	12.299	0.777

S10 Quantifier error

Table S10 Mean absolute error of prevalence estimators by batch size (the row guard_fire_rate is the guard's rejection rate)

quantifier	100	500	2000	10000
guard_fire_rate	0.062	0.070	0.074	0.104
pi_acc	0.058	0.030	0.018	0.013
pi_cc	0.079	0.076	0.076	0.075
pi_em	0.047	0.023	0.014	0.010
pi_hdy	0.057	0.026	0.016	0.012
pi_mean	0.041	0.022	0.014	0.011
pi_mle	0.048	0.024	0.015	0.012
pi_pacc	0.048	0.025	0.015	0.011
pi_pcc	0.081	0.080	0.079	0.079

S11 Ablation: paired tests

Table S11 Paired tests of PAFT-Bayes against ablated variants (all batch sizes)

baseline	n_units	mean_diff	ci95_lo	ci95_hi	wins	ties	losses	p_wilcoxon	rank_biserial	p_holm
paft_plug	18	+0.0102	+0.0071	+0.0132	18	0	0	0.00001	1.000	0.00007
paft_guard	18	+0.0006	−0.0001	+0.0013	13	1	4	0.01682	0.660	0.06372
bayes_sd0.5	18	−0.0005	−0.0031	+0.0018	9	0	9	0.79871	0.076	0.79871
bayes_sd1.0	18	−0.0020	−0.0036	−0.0007	6	0	12	0.01593	−0.637	0.06372
bayes_sd4.0	18	+0.0036	+0.0018	+0.0057	18	0	0	0.00001	1.000	0.00007
bayes_sdNone	18	+0.0071	+0.0035	+0.0117	18	0	0	0.00001	1.000	0.00007
plug_acc	18	+0.0108	+0.0063	+0.0149	16	0	2	0.00025	0.895	0.00151
plug_em	18	+0.0067	+0.0040	+0.0092	17	0	1	0.00025	0.895	0.00151
dta_hist	18	+0.0017	−0.0028	+0.0049	13	0	5	0.04317	0.544	0.08633

S12 Other metrics: paired tests

Table S12 Paired tests for F0.5, F2 and MCC

metric	baseline	n_units	mean_diff	ci95_lo	ci95_hi	wins	ties	losses	p_wilcoxon	rank_biserial	p_holm
f0.5	static	18	+0.0259	+0.0079	+0.0379	17	0	1	0.00193	0.789	0.00386
f0.5	dta_em	18	−0.0074	−0.0224	+0.0008	10	0	8	0.89858	0.041	0.89858
f0.5	em_fixed	18	+0.0356	+0.0243	+0.0471	17	0	1	0.00002	0.977	0.00007
f2	static	18	+0.0397	+0.0282	+0.0521	17	0	1	0.00002	0.977	0.00005
f2	dta_em	18	+0.0007	−0.0032	+0.0034	12	0	6	0.08977	0.462	0.08977
f2	em_fixed	18	+0.0193	+0.0129	+0.0264	18	0	0	0.00001	1.000	0.00002
mcc	static	18	+0.0144	+0.0079	+0.0217	15	0	3	0.00025	0.895	0.00025
mcc	em_fixed	18	+0.0400	+0.0242	+0.0585	18	0	0	0.00001	1.000	0.00002

S13 Label-assisted variant: all budgets and paired tests

Table S13 Label-assisted adaptation for all label budgets

suite	n_lab	Static-F	PAFT-Bayes (no labels)	Re-tune on labels	PAFT-L (labels → π)	PAFT-L + mixture	PAFT-L guarded	Oracle-π	Oracle-threshold
acs_income_states	50	0.702	0.693	0.692	0.712	0.693	0.711	0.713	0.720
acs_income_states	200	0.702	0.693	0.709	0.713	0.694	0.712	0.713	0.720
acs_income_states	1,000	0.702	0.693	0.715	0.713	0.697	0.712	0.713	0.720
bank_time	50	0.233	0.265	0.237	0.259	0.265	0.259	0.260	0.272
bank_time	200	0.233	0.265	0.248	0.259	0.264	0.259	0.259	0.272
bank_time	1,000	0.233	0.265	0.263	0.260	0.261	0.260	0.260	0.275
controlled	50	0.498	0.535	0.490	0.532	0.536	0.534	0.538	0.564
controlled	200	0.497	0.535	0.506	0.535	0.536	0.535	0.538	0.565
creditcard_time	50	0.645	0.640	0.625	0.649	0.640	0.649	0.646	0.729
creditcard_time	200	0.646	0.640	0.585	0.658	0.641	0.658	0.647	0.730
creditcard_time	1,000	0.644	0.641	0.514	0.639	0.639	0.639	0.647	0.732
electricity_time	50	0.450	0.426	0.692	0.449	0.430	0.449	0.450	0.721
electricity_time	200	0.450	0.426	0.706	0.449	0.436	0.449	0.450	0.722
elliptic_time	50	0.483	0.477	0.486	0.486	0.477	0.486	0.480	0.602
elliptic_time	200	0.484	0.477	0.543	0.481	0.479	0.481	0.485	0.604
elliptic_time	1,000	0.863	0.864	0.875	0.863	0.864	0.863	0.863	0.883
kdd99_time	50	0.973	0.947	0.964	0.951	0.947	0.951	0.950	0.988
kdd99_time	200	0.973	0.947	0.981	0.951	0.947	0.951	0.950	0.989
kdd99_time	1,000	0.973	0.947	0.987	0.950	0.946	0.950	0.950	0.989

S13 (continued)

Table S13b Paired tests of guarded PAFT-L

suite	n_lab	baseline	n_units	mean_diff	ci95_lo	ci95_hi	wins	ties	losses	p_wilcoxon	rank_biserial	p_holm
acs_income_states	50	static	50	+0.0087	+0.0052	+0.0133	39	0	11	0.0000	0.743	0.0000
acs_income_states	50	retune_labels	50	+0.0192	+0.0171	+0.0214	50	0	0	0.0000	1.000	0.0000
acs_income_states	50	paft_bayes	50	+0.0180	+0.0148	+0.0215	47	1	2	0.0000	0.990	0.0000
acs_income_states	200	static	50	+0.0097	+0.0061	+0.0145	40	0	10	0.0000	0.795	0.0000
acs_income_states	200	retune_labels	50	+0.0032	+0.0019	+0.0045	39	0	11	0.0000	0.666	0.0000
acs_income_states	200	paft_bayes	50	+0.0190	+0.0157	+0.0225	49	1	0	0.0000	1.000	0.0000
acs_income_states	1,000	static	50	+0.0100	+0.0064	+0.0149	39	0	11	0.0000	0.804	0.0000
acs_income_states	1,000	retune_labels	50	−0.0030	−0.0041	−0.0019	14	0	36	0.0000	−0.667	0.0000
acs_income_states	1,000	paft_bayes	50	+0.0192	+0.0160	+0.0228	50	0	0	0.0000	1.000	0.0000
bank_time	50	static	10	+0.0258	+0.0078	+0.0497	9	0	1	0.0059	0.927	0.0176
bank_time	50	retune_labels	10	+0.0223	+0.0113	+0.0328	9	0	1	0.0137	0.855	0.0273
bank_time	50	paft_bayes	10	−0.0063	−0.0167	+0.0004	2	0	8	0.1934	−0.491	0.1934
bank_time	200	static	10	+0.0258	+0.0075	+0.0499	8	0	2	0.0098	0.891	0.0293
bank_time	200	retune_labels	10	+0.0101	+0.0020	+0.0166	8	0	2	0.0645	0.673	0.1289
bank_time	200	paft_bayes	10	−0.0062	−0.0169	+0.0009	3	0	7	0.2754	−0.418	0.2754
bank_time	1,000	static	10	+0.0266	+0.0084	+0.0512	9	0	1	0.0059	0.927	0.0176
bank_time	1,000	retune_labels	10	−0.0029	−0.0134	+0.0039	6	0	4	0.6953	0.164	0.6953
bank_time	1,000	paft_bayes	10	−0.0055	−0.0156	+0.0013	3	0	7	0.2754	−0.418	0.5508
controlled	50	static	18	+0.0361	+0.0266	+0.0461	17	0	1	0.0000	0.988	0.0000
controlled	50	retune_labels	18	+0.0445	+0.0353	+0.0544	18	0	0	0.0000	1.000	0.0000
controlled	50	paft_bayes	18	−0.0007	−0.0037	+0.0010	12	0	6	0.0814	0.474	0.0814
controlled	200	static	18	+0.0378	+0.0294	+0.0472	18	0	0	0.0000	1.000	0.0000
controlled	200	retune_labels	18	+0.0287	+0.0197	+0.0391	18	0	0	0.0000	1.000	0.0000
controlled	200	paft_bayes	18	+0.0007	−0.0003	+0.0014	17	0	1	0.0019	0.789	0.0019
creditcard_time	50	static	14	+0.0045	−0.0136	+0.0224	9	0	5	0.4631	0.238	0.4631
creditcard_time	50	retune_labels	14	+0.0241	+0.0038	+0.0440	11	0	3	0.0245	0.676	0.0736
creditcard_time	50	paft_bayes	14	+0.0091	+0.0001	+0.0183	9	0	5	0.0785	0.543	0.1570
creditcard_time	200	static	14	+0.0123	−0.0048	+0.0321	7	0	7	0.5016	0.219	0.5016
creditcard_time	200	retune_labels	14	+0.0731	+0.0483	+0.0960	12	0	2	0.0006	0.943	0.0018
creditcard_time	200	paft_bayes	14	+0.0181	+0.0079	+0.0284	11	0	3	0.0067	0.790	0.0134
creditcard_time	1,000	static	14	−0.0052	−0.0213	+0.0113	5	0	9	0.4263	−0.257	0.8525
creditcard_time	1,000	retune_labels	14	+0.1250	+0.0934	+0.1530	13	0	1	0.0002	0.981	0.0007
creditcard_time	1,000	paft_bayes	14	−0.0021	−0.0131	+0.0081	8	0	6	0.9515	−0.029	0.9515
electricity_time	50	static	31	−0.0013	−0.0026	−0.0001	11	1	19	0.0859	−0.359	0.0859
electricity_time	50	retune_labels	31	−0.2429	−0.3180	−0.1674	7	0	24	0.0000	−0.887	0.0000
electricity_time	50	paft_bayes	31	+0.0231	+0.0150	+0.0321	25	0	6	0.0000	0.903	0.0000
electricity_time	200	static	31	−0.0008	−0.0020	+0.0002	10	5	16	0.1587	−0.316	0.1587
electricity_time	200	retune_labels	31	−0.2565	−0.3323	−0.1802	5	0	26	0.0000	−0.919	0.0000
electricity_time	200	paft_bayes	31	+0.0234	+0.0152	+0.0325	26	0	5	0.0000	0.915	0.0000
elliptic_time	50	static	15	+0.0031	−0.0049	+0.0161	4	4	7	0.2132	−0.424	0.6397
elliptic_time	50	retune_labels	15	−0.0005	−0.0950	+0.0786	9	0	6	0.3894	0.267	0.7788
elliptic_time	50	paft_bayes	15	+0.0090	−0.0015	+0.0267	5	4	6	1.0000	0.000	1.0000
elliptic_time	200	static	15	−0.0023	−0.0218	+0.0163	5	4	6	0.5937	−0.182	1.0000
elliptic_time	200	retune_labels	15	−0.0614	−0.1523	+0.0119	8	0	7	0.5614	−0.183	1.0000
elliptic_time	200	paft_bayes	15	+0.0042	−0.0001	+0.0094	6	4	5	0.3281	0.333	0.9842
elliptic_time	1,000	static	1	+0.0004	+0.0004	+0.0004	1	0	0	1.0000	1.000	1.0000
elliptic_time	1,000	retune_labels	1	−0.0115	−0.0115	−0.0115	0	0	1	1.0000	−1.000	1.0000
elliptic_time	1,000	paft_bayes	1	−0.0003	−0.0003	−0.0003	0	0	1	1.0000	−1.000	1.0000
kdd99_time	50	static	14	−0.0220	−0.0426	−0.0070	1	0	13	0.0004	−0.962	0.0011
kdd99_time	50	retune_labels	14	−0.0128	−0.0256	−0.0011	4	0	10	0.0676	−0.562	0.1353
kdd99_time	50	paft_bayes	14	+0.0043	−0.0087	+0.0162	11	0	3	0.0785	0.543	0.1353
kdd99_time	200	static	14	−0.0225	−0.0434	−0.0067	2	0	12	0.0009	−0.924	0.0026
kdd99_time	200	retune_labels	14	−0.0303	−0.0568	−0.0106	4	0	10	0.0107	−0.752	0.0215
kdd99_time	200	paft_bayes	14	+0.0039	−0.0094	+0.0156	9	2	3	0.1361	0.487	0.1361
kdd99_time	1,000	static	14	−0.0238	−0.0481	−0.0060	2	0	12	0.0009	−0.924	0.0026
kdd99_time	1,000	retune_labels	14	−0.0370	−0.0742	−0.0115	3	0	11	0.0023	−0.867	0.0046
kdd99_time	1,000	paft_bayes	14	+0.0025	−0.0103	+0.0145	8	3	3	0.2860	0.364	0.2860

S14 Per-dataset results

Table S14 Mean F1 under shift per dataset (LightGBM)

dataset	PAFT-Bayes	Static-F	DTA-EM	EM-Fixed	Oracle-π	Oracle-threshold
APSFailure	0.763	0.735	0.771	0.754	0.772	0.832
Amazon_employee	0.358	0.322	0.355	0.314	0.364	0.401
Click_small	0.409	0.372	0.405	0.368	0.415	0.433
Diabetes130US	0.331	0.284	0.329	0.307	0.335	0.357
KDD09_appetency	0.189	0.173	0.187	0.161	0.196	0.243
KDD09_churn	0.271	0.253	0.264	0.232	0.272	0.296
KDD09_upselling	0.537	0.514	0.537	0.523	0.540	0.567
MiniBooNE	0.869	0.840	0.869	0.866	0.870	0.879
adult	0.709	0.647	0.709	0.704	0.711	0.727
bank_marketing	0.592	0.560	0.591	0.560	0.594	0.618
census_income	0.600	0.548	0.599	0.582	0.605	0.639
creditcard	0.749	0.747	0.790	0.677	0.770	0.837
default_credit	0.539	0.486	0.537	0.501	0.542	0.559
jm1	0.448	0.406	0.447	0.422	0.455	0.478
kick	0.480	0.391	0.480	0.470	0.483	0.507
nomao	0.932	0.912	0.932	0.932	0.932	0.940
online_shoppers	0.638	0.610	0.639	0.610	0.644	0.671
porto_seguro	0.163	0.138	0.160	0.134	0.167	0.193

S15 Full efficiency measurements

Table S15 All latency and memory measurements

component	m	ms	peak_mb
fit decision layer (once, incl. 50 bootstrap static thresholds)	9,766	3696.230	14.19
static threshold (comparison only)	100	0.001	–
EM prior correction (SLD)	100	0.149	–
DTA-EM (EM + sort + expected-F scan)	100	0.166	–
PAFT-Plug	100	0.036	–
PAFT-Bayes	100	0.429	–
PAFT guard (KS test)	100	0.066	–
all rules (decide)	100	1.149	–
static threshold (comparison only)	500	0.003	–
EM prior correction (SLD)	500	0.186	–
DTA-EM (EM + sort + expected-F scan)	500	0.184	–
PAFT-Plug	500	0.042	–
PAFT-Bayes	500	0.498	–
PAFT guard (KS test)	500	0.089	–
all rules (decide)	500	1.283	–
static threshold (comparison only)	2,000	0.001	–
EM prior correction (SLD)	2,000	0.270	–
DTA-EM (EM + sort + expected-F scan)	2,000	0.281	–
PAFT-Plug	2,000	0.081	–
PAFT-Bayes	2,000	0.558	–
PAFT guard (KS test)	2,000	0.194	–
all rules (decide)	2,000	1.513	–
static threshold (comparison only)	10,000	0.001	–
EM prior correction (SLD)	10,000	0.727	–
DTA-EM (EM + sort + expected-F scan)	10,000	0.886	–
PAFT-Plug	10,000	0.322	–
PAFT-Bayes	10,000	0.765	–
PAFT guard (KS test)	10,000	0.567	–
all rules (decide)	10,000	3.040	–
static threshold (comparison only)	100,000	0.009	–
EM prior correction (SLD)	100,000	13.237	–
DTA-EM (EM + sort + expected-F scan)	100,000	10.183	–
PAFT-Plug	100,000	2.852	–
PAFT-Bayes	100,000	3.162	–
PAFT guard (KS test)	100,000	4.333	–
all rules (decide)	100,000	29.978	–
retrain LightGBM on adult (reference)	24,424	1511.210	–
stored state: thresholds T x grid G surface (float64)	329,728	–	2.52

S16 Guard calibration on synthetic scores (Experiment 10)

Table S16 Rejection rate of the guard (level 0.01) on binormal scores (class separation 2 s.d., development prevalence 0.05) when every batch is a fresh sample. score_shift = 0 is pure label shift (rate = size); score_shift > 0 displaces all deployment scores by that many s.d. (rate = power). 500 batches per cell over 10 validation draws

score_shift	n1_val	rho=1, m=500	rho=1, m=2000	rho=1, m=10000	rho=4, m=500	rho=4, m=2000	rho=4, m=10000
0.000	50.000	0.010	0.002	0.068	0.000	0.004	0.000
0.000	100.000	0.000	0.002	0.000	0.000	0.000	0.002
0.000	500.000	0.004	0.002	0.002	0.000	0.000	0.002
0.000	5000.000	0.002	0.002	0.004	0.000	0.000	0.006
0.250	50.000	0.352	0.610	0.990	0.028	0.138	0.180
0.250	100.000	0.486	0.826	0.992	0.048	0.358	0.226
0.250	500.000	0.604	0.996	1.000	0.120	0.718	1.000
0.250	5000.000	0.618	0.998	1.000	0.142	0.932	1.000
0.500	50.000	0.698	0.830	0.920	0.080	0.138	0.166
0.500	100.000	0.854	1.000	1.000	0.138	0.412	0.670
0.500	500.000	0.988	1.000	1.000	0.396	0.996	1.000
0.500	5000.000	0.994	1.000	1.000	0.614	1.000	1.000

S17 Dataset screening

Table S17 OpenML datasets screened against the inclusion criteria (binary target, at least 10,000 instances, at least 400 minority instances, minority prevalence at most 30 %)

Dataset	OpenML id	Decision
adult	1,590	included
bank-marketing	1,461	included
creditcard	1,597	included
default-of-credit-card-clients	42,477	included
APSFailure	41,138	included
KDDCup09_appetency	1,111	included
KDDCup09_churn	1,112	included
KDDCup09_upselling	1,114	included
Click_prediction_small	1,220	included
nomao	1,486	included
MiniBooNE	41,150	included
jm1	1,053	included
Amazon_employee_access	4,135	included
kick	41,162	included
Diabetes130US	4,541	included (binarised: readmission < 30 days)
online_shoppers	45,060	included
porto-seguro	42,742	included
Census-Income-KDD	42,750	included
MagicTelescope	1,120	excluded: minority prevalence 35 %
eeg-eye-state	1,471	excluded: minority prevalence 45 %
electricity	151	excluded: minority prevalence 42 % (used in the temporal suite)
PhishingWebsites	4,534	excluded: minority prevalence 44 %
airlines	1,169	excluded: minority prevalence 45 %
higgs	23,512	excluded: minority prevalence 47 %
numerai28.6	23,517	excluded: minority prevalence 49 %
mammography	310	excluded: 260 minority instances
road-safety	42,803	excluded: target not binary
data id 45557	45,557	excluded: fewer than 10,000 instances
SantanderCustomerSatisfaction	42,395	eligible by description but download failed repeatedly; not included

S18 Sensitivity to cells with few expected positives

Table S18 Paired tests of PAFT-Bayes restricted to benchmark cells whose expected number of positives (target prevalence times batch size) exceeds a minimum

cells	share_of_cells	baseline	n_units	mean_diff	ci95_lo	ci95_hi	wins	ties	losses	p_wilcoxon	rank_biserial	p_holm
all cells	1.00	static	18	+0.0355	+0.0270	+0.0453	18	0	0	0.00001	1.000	0.00002
all cells	1.00	dta_em	18	−0.0014	−0.0067	+0.0020	14	0	4	0.09874	0.450	0.09874
all cells	1.00	em_fixed	18	+0.0255	+0.0178	+0.0337	18	0	0	0.00001	1.000	0.00002
expected positives >= 5	0.93	static	18	+0.0376	+0.0302	+0.0463	18	0	0	0.00001	1.000	0.00002
expected positives >= 5	0.93	dta_em	18	+0.0012	+0.0003	+0.0022	15	0	3	0.00896	0.684	0.00896
expected positives >= 5	0.93	em_fixed	18	+0.0230	+0.0165	+0.0294	18	0	0	0.00001	1.000	0.00002
expected positives >= 20	0.81	static	18	+0.0401	+0.0333	+0.0482	18	0	0	0.00001	1.000	0.00002
expected positives >= 20	0.81	dta_em	18	+0.0007	+0.0001	+0.0015	11	0	7	0.28373	0.298	0.28373
expected positives >= 20	0.81	em_fixed	18	+0.0199	+0.0139	+0.0258	18	0	0	0.00001	1.000	0.00002

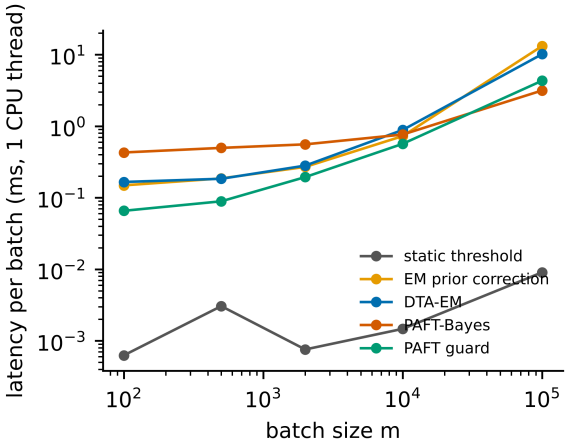


Fig. S1 Latency of the decision rules as a function of batch size (one CPU thread)